

AI ASSISTED CODING

LAB - 7.5

M.Saikrishna

2303A51771

BATCH-12

Task 1 (Mutable Default Argument – Function Bug)

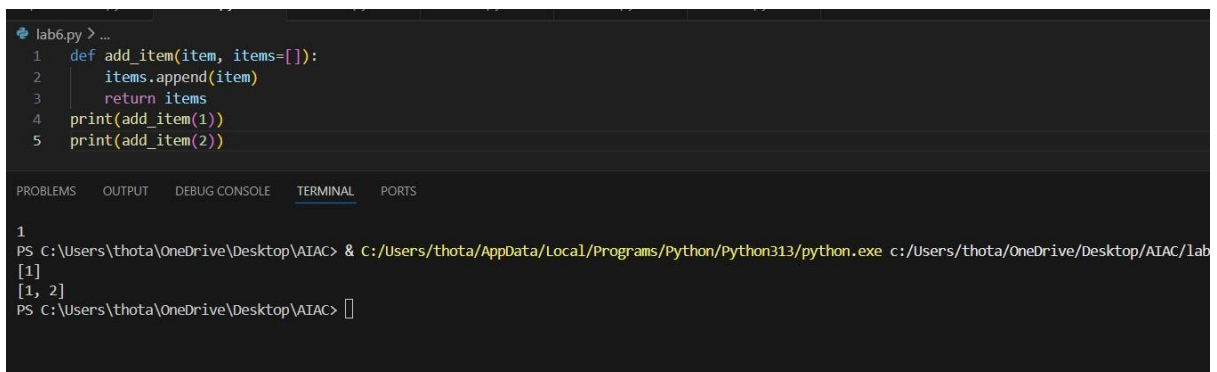
Task: Analyze given code where a mutable default argument causes unexpected behavior. Use AI to fix it. # Bug: Mutable default

argument

`def add_item(item, items=[]):`
`items.append(item)` `return`

`items` `print(add_item(1))` `print(add_item(2))`

Expected Output: Corrected function avoids shared list bug.



```
lab6.py > ...
1 def add_item(item, items=[]):
2     items.append(item)
3     return items
4 print(add_item(1))
5 print(add_item(2))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

1
PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
[1]
[1, 2]
PS C:\Users\thota\OneDrive\Desktop\AIAC>
```

Task 2 (Floating-Point Precision Error)

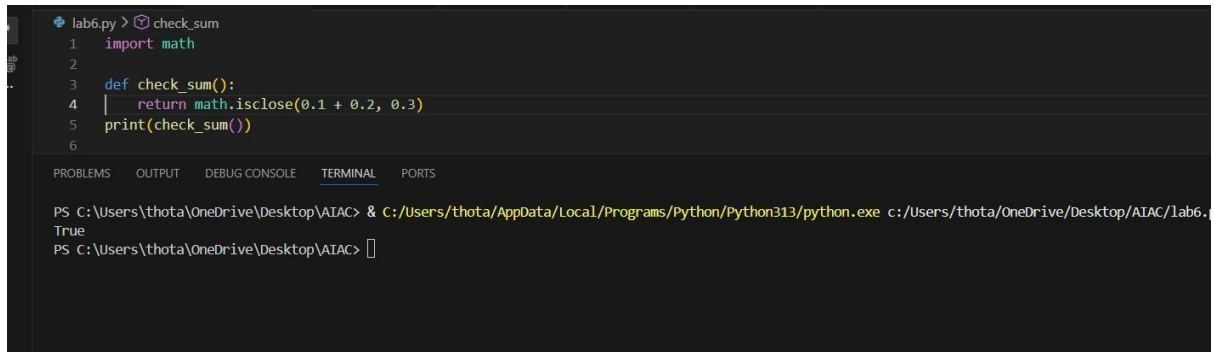
Task: Analyze given code where floating-point comparison fails.

Use AI to correct with tolerance. # Bug: Floating point precision issue

```
def check_sum(): return (0.1 + 0.2)
```

```
== 0.3 print(check_sum())
```

Expected Output: Corrected function



The screenshot shows a code editor with a file named 'lab6.py'. The code defines a function 'check_sum()' that returns the sum of 0.1 and 0.2, and then prints the result. The terminal output shows the command to run the script, followed by the output 'True'.

```
lab6.py > check_sum
1 import math
2
3 def check_sum():
4     return math.isclose(0.1 + 0.2, 0.3)
5 print(check_sum())
6

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
True
PS C:\Users\thota\OneDrive\Desktop\AIAC>
```

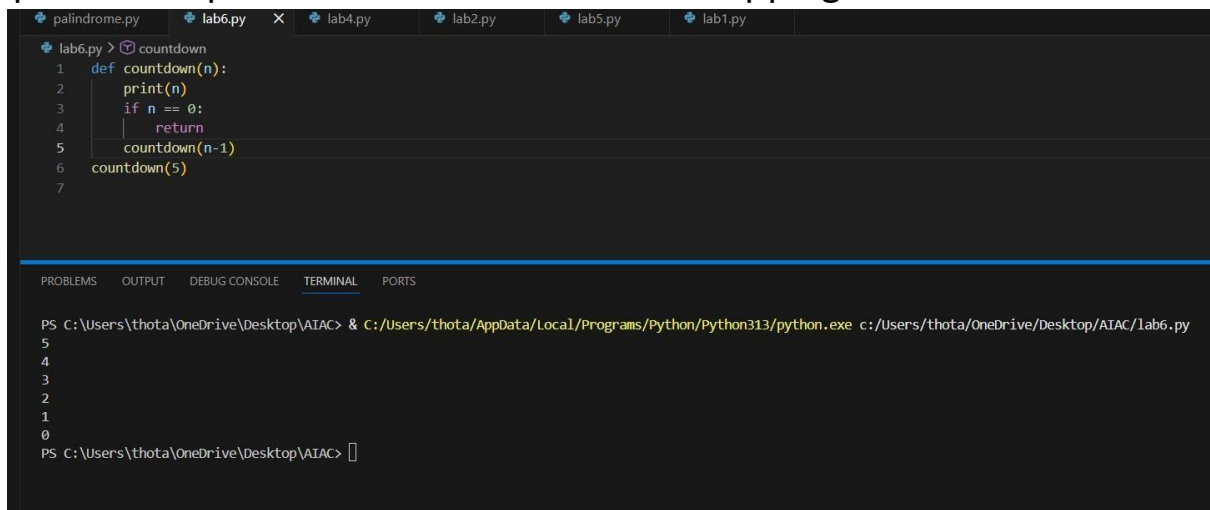
Task 3 (Recursion Error – Missing Base Case)

Task: Analyze given code where recursion runs infinitely due to missing base case. Use AI to fix. # Bug: No base case def

```
countdown(n): print(n)
```

```
return countdown(n-1) countdown(5)
```

Expected Output : Correct recursion with stopping condition



The screenshot shows a code editor with a file named 'lab6.py'. The code defines a recursive function 'countdown(n)' that prints the value of 'n' and then calls itself with 'n-1'. The terminal output shows the command to run the script, followed by the output '5 4 3 2 1 0'.

```
lab6.py > countdown
1 def countdown(n):
2     print(n)
3     if n == 0:
4         return
5     countdown(n-1)
6     countdown(5)
7

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
5
4
3
2
1
0
PS C:\Users\thota\OneDrive\Desktop\AIAC>
```

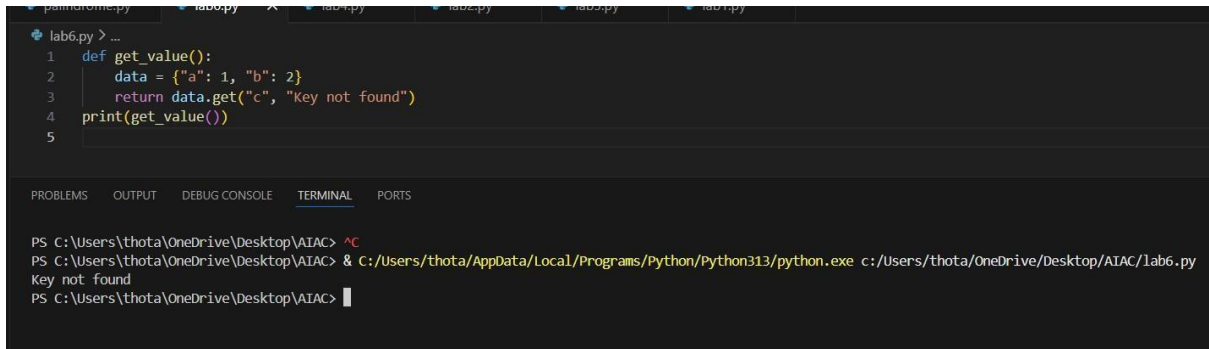
Task 4 (Dictionary Key Error)

Task: Analyze given code where a missing dictionary key causes error. Use AI to fix it.

Bug: Accessing non-existing key

```
def get_value(): data = {"a": 1, "b":  
2} return data["c"]  
print(get_value())
```

Expected Output: Corrected with `.get()` or error handling.



```
lab6.py > ...  
1 def get_value():  
2     data = {"a": 1, "b": 2}  
3     return data.get("c", "Key not found")  
4 print(get_value())  
5  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS C:\Users\thota\OneDrive\Desktop\AIAC> ^C  
PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py  
Key not found  
PS C:\Users\thota\OneDrive\Desktop\AIAC> |
```

Task 5 (Infinite Loop – Wrong Condition)

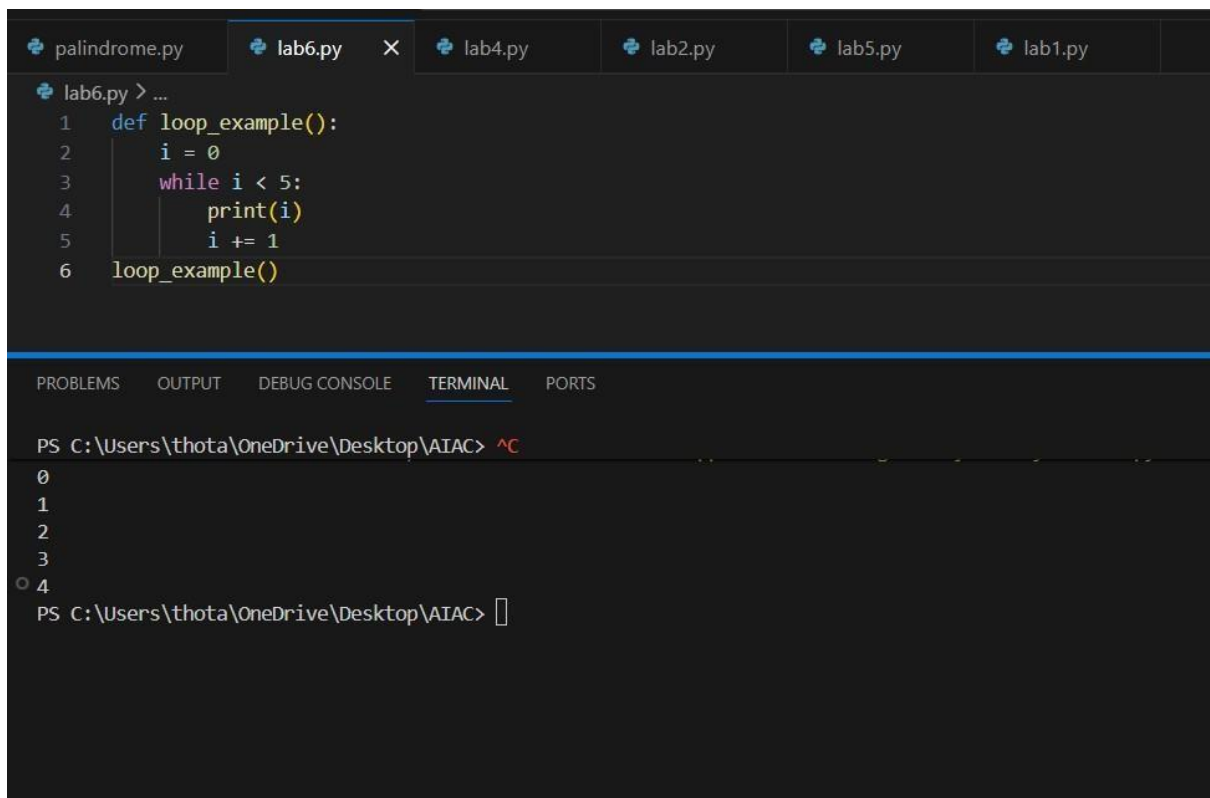
Task: Analyze given code where loop never ends. Use AI to detect and fix it.

Bug: Infinite loop def loop_example():

i = 0 while

i < 5: print(i)

Expected Output: Corrected loop increments i.



The screenshot shows a code editor with several tabs: `palindrome.py`, `lab6.py` (active), `lab4.py`, `lab2.py`, `lab5.py`, and `lab1.py`. The `lab6.py` tab contains the following Python code:

```
1 def loop_example():
2     i = 0
3     while i < 5:
4         print(i)
5         i += 1
6 loop_example()
```

Below the code editor is a terminal window with tabs for `PROBLEMS`, `OUTPUT`, `DEBUG CONSOLE`, `TERMINAL` (active), and `PORTS`. The terminal shows the command prompt `PS C:\Users\thota\OneDrive\Desktop\AIAC>` followed by the output of the `loop_example()` function:

```
0
1
2
3
4
PS C:\Users\thota\OneDrive\Desktop\AIAC>
```

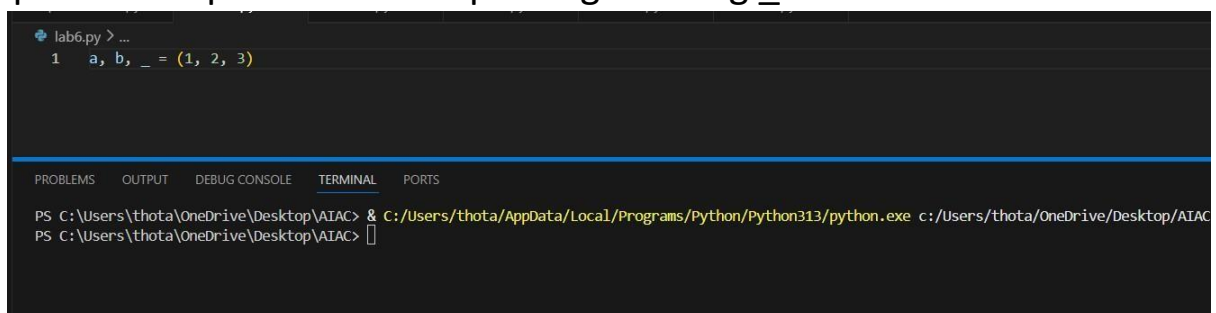
Task 6 (Unpacking Error – Wrong Variables)

Task: Analyze given code where tuple unpacking fails. Use AI to fix it.

Bug: Wrong unpacking

`a, b = (1, 2, 3)`

Expected Output: Correct unpacking or using `_` for extra values.



The screenshot shows a code editor with a single tab `lab6.py` containing the following Python code:

```
1 a, b, _ = (1, 2, 3)
```

Below the code editor is a terminal window with tabs for `PROBLEMS`, `OUTPUT`, `DEBUG CONSOLE`, `TERMINAL` (active), and `PORTS`. The terminal shows the command prompt `PS C:\Users\thota\OneDrive\Desktop\AIAC>` followed by the command `C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC` and the output:

```
PS C:\Users\thota\OneDrive\Desktop\AIAC>
```

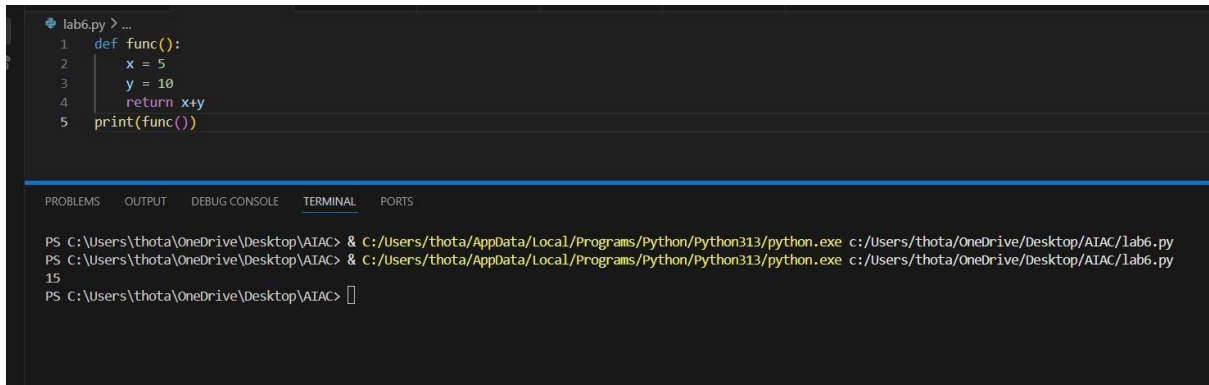
Task 7 (Mixed Indentation – Tabs vs Spaces)

Task: Analyze given code where mixed indentation breaks

execution. Use AI to fix it. # Bug: Mixed indentation def

func(): x = 5 y = 10 return x+y

Expected Output : Consistent indentation applied.



```
lab6.py > ...
1 def func():
2     x = 5
3     y = 10
4     return x+y
5 print(func())
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
15
PS C:\Users\thota\OneDrive\Desktop\AIAC>
```

Task 8 (Import Error – Wrong Module Usage)

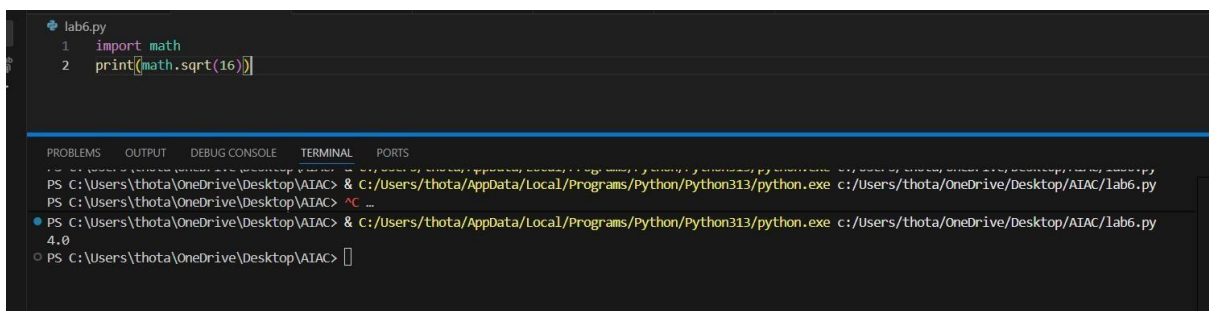
Task: Analyze given code with incorrect import. Use AI to fix.

Bug: Wrong import import

maths print(maths.sqrt(16))

Expected Output: Corrected to

import math



```
lab6.py
1 import maths
2 print(maths.sqrt(16))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
PS C:\Users\thota\OneDrive\Desktop\AIAC> ^C ...
PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
4.0
PS C:\Users\thota\OneDrive\Desktop\AIAC>
```

